

SHANGYANG MIN

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EDUCATION

Brown University

Sep 2023 - May 2025

Sc.M. in Computer Science | GPA: 4.0/4.0 | Multimodal AI, Brain-Inspired Computing

Michigan State University

Aug 2019 - May 2023

B.S. in Computer Science Engineering, High Honor | GPA: 3.95/4.0 | ML, Computer Vision, Biomedical Imaging

EXPERIENCE

Hubei Lianhuashan AI Research Institute

Jun 2025 - Present

Research Engineer | Under Peking University Institute for General AI | Ezhou, China

- Designed and taught an open-source LLM & Agent curriculum with runnable notebooks, spanning model fundamentals to agent systems; delivered through executive workshops, technical bootcamps, and university courses.
- Drove applied AI initiatives with enterprise partners across Hubei, taking projects end-to-end from requirements and solution design to prototype and deployment across traditional industries.

Abysen AI

Feb 2025 - Present

Founder & AI Engineer (Part-time) | Remote | abysen.com

- Design and build applied AI products end-to-end, from agent frameworks to real-time voice agents, shipped as open-source systems and a mobile app.

Lee Lab, Brown University

Sep 2024 - May 2025

Research Assistant | Brain-Computer Interface and Multimodal Interaction

- Built a real-time EEG-to-Unity avatar-control pipeline and developed a vision-EEG fusion architecture for cross-modal interaction.

HAAI Lab, Michigan State University

May 2022 - Aug 2023

Research Assistant | Biomedical Imaging | NSF and HFH+MSU Cancer Grant

- Proposed Feature Imitating Networks (FINs), which embed closed-form radiomics features into lightweight networks for CT/MRI classification and segmentation, improving accuracy while cutting inference time 3 $\times$.

Henry Ford Health x Michigan State University

Sep 2022 - Sep 2023

Research Assistant | NIH-Funded Program | Tumor Detection and Radiomics

- Built DCE-MRI tumor-detection and radiomics pipelines; standardized preprocessing for multi-institutional model evaluation.

SELECTED PROJECTS

Aezab: Self-Hosted Agent Framework

- Built a self-hosted agent framework where agents are assembled and invoked through conversation — knowledge bases, workflows, tools, and delegated sub-agents exposed as callable capabilities — with hybrid retrieval, streaming APIs, and end-to-end tracing.

Tethro: Real-Time Personal Voice Agent

- Engineered a dual-loop architecture that decouples the real-time voice interface from a text-based reasoning core: the voice loop keeps conversation fluent and interruptible while the reasoning loop guarantees reliable answers and actions, achieving sub-second voice-to-voice response; includes long-term memory and user-confirmed actions.

Trigger-Simulation Backdoor Removal Defense (TSBRD)

- Proposed virtual-prompt realignment for backdoored LLMs, reducing attack success rate from 93% to 0% on Llama-2 within 7 epochs while preserving MMLU accuracy.

VideoQA with External Vision Models

- Designed a modular VideoQA pipeline that converts video into natural-language evidence via external vision models before question answering — matched fine-tuned SeViLA-level accuracy with zero-shot GPT-4, without task-specific video training.

Additional research projects at my [project page](#).

TECHNICAL SKILLS

Languages: Python, TypeScript, C, C++, SQL

Machine Learning: PyTorch, TensorFlow, Hugging Face (Transformers, PEFT, TRL), vLLM, Weights & Biases

Engineering: FastAPI, React, React Native, Docker, PostgreSQL/pgvector, Redis, Git, Linux

Research Areas: LLMs and agent systems, multimodal learning, computer vision, biomedical imaging, real-time voice AI

PUBLICATION

S. Min, H. B. Ebadian, T. Alhanai, M. M. Ghassemi, "Feature Imitating Networks Enhance the Performance, Reliability and Speed of Deep Learning on Biomedical Image Processing Tasks," IEEE EMBC 2024.